University of Calgary

Schulich School of Engineering

ENEL 678 - Graduate Project in Electrical Engineering

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Final Report

Topic:

Design and Optimization of a Smart EV Charging Station with Integrated Solar PV and Electrical Grid

Team Number: 5

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1| Glossary

BESS – Battery Energy Storage System; BMS – Battery Management System; CS – Charging Station; DCFC – DC Fast Charging; EVs – Electric Vehicles; Isc – Short Circuit Current; MPPT – Maximum Power Point Tracking; PV – Photovoltaic; ROI – Return of Investment; SOC – State of Charging; OCPP – Open Charge Point Protocol; DOD – Depth of Discharge; Voc – Open Circuit Voltage; ,EMS-Energy management system.

Keywords *– Battery, Electric Vehicles (EVs), PV System, EV Charging Station, Optimal Sizing.*

2| Executive Summary

The ***Design and Optimization of a Smart EV Charging Station with Integrated Solar PV and Electrical Grid*** *project, led by Team 5 at the University of Calgary addresses* the main goal is to support sustainable transportation in remote areas by designing and optimizing a smart EV charging station that is connected to the electrical grid, solar photovoltaic (PV) panels, and battery energy storage. Through thorough load analysis and the creation of comprehensive system specifications for essential parts like solar modules, batteries, and DC/DC converters, the team has successfully designed a system that effectively converts 208V three-phase utility power into a stable 400V DC bus. While issues with voltage ripple, converter selection, grid synchronization, and charging circuit stability have been resolved through iterative design improvements and reliable control algorithms, a fuzzy logic-based MPPT controller that optimizes power output under varying conditions has been developed because of sophisticated simulations in MATLAB and Simulink. Furthermore, strategic financial planning and a well-defined $1.5 million budget guarantee the project's feasibility and return on investment. Future improvements are anticipated to further increase sustainability and energy efficiency, including induction charging and vehicle-to-grid integration. This initiative establishes the foundation for a scalable EV charging system that satisfies the need for contemporary mobility infrastructure, with a particular emphasis on dependability, optimization, and integration of renewable energy.

3| Project Motivation and Objectives

The shift to electric vehicles (EVs) is crucial for decarbonising the transportation industry, which contributes to significant amount of global CO_2 emissions. EVs have a lot of potential to cut greenhouse gas emissions, but when they are charged from grids that rely on fossil fuels, their environmental benefits are severely diminished.

The problem is worsened by a lack of charging facilities, especially in countryside regions. For example, Jasper National Park, a well-known tourist site, receives more than two million visitors each year, and as of now, there are fewer DC fast charging (DCFC) outlets located within a distance. The customers of EVs suffer from severe “range anxiety” due to this gap in infrastructure, which further increases the strain on the local power systems. There is a possibility that the grid’s reliability will be at risk if the high-powered charging stations located in these off-grid areas experience voltage sags.

The project scope tries to achieve the goals by building and operationalizing a smart solar-enabled EV fast charging station and aims to accomplish the following specific milestones.

The installation of 600 kW solar photovoltaic with a supportive operational and energy autonomy 550 kWh battery storage to enhance energy autonomy will also be added. Five 150 kW charging station will be added to the current setup of fast chargers with the capability of 20-80% SoC in 20-30 min. WE would also implement a smart energy management system (EMS) to optimize battery charging and discharging from renewables and the grid which strengthens power flow management.

4| Methodology and Design

The methodology that has been followed to reach the final solution and outcome is determining the project purpose at the early stage and adapting of project management to comply with requirements and validating the design at different stages through detailed analysis and searching for alternative solutions to reach optimization and efficiency.

Table 1: Methodology Steps

Step No.	**Methodology Steps**	**Descriptions**
Step 01	Analyzing the current problem	The challenges in the adaptation of EVs at a large scale are due to a lack of Fast charging stations in remote areas.
Step 02	Proposing a solution	Design of a DC fast charging station with renewable and grid integration to ensure green energy and reliability simultaneously.

Step 03	Layout of system	Outlining the architecture and structure of the proposed system with EV, grid, BESS, loads, and microgrid.
Step 04	Defining specification	Based on load demand, source availability and efficient approach specifying components to reach desired output.
Step 05	Designing the system components	Design of solar panels, Boost converters, Buck boost converters, inverters, rectifiers to reach the final load and voltage.
Step 06	Optimizing the system	Integration of MPPT in solar, Battery management system for BESS, Energy management system for load balancing has been implemented
Step 07	Approaching different methodologies	Different methodology to reach the best fit like different algorithm for MPPT, BMS system and EMS systems have been experimented through simulation to reach a stable outcome
Step 08	Simulation and testing	All the integrated part has been simulated in MATLAB Simulink/ETAP at different stages of the project
Step 09	Test and validation	Compare the simulation output at different conditions like irradiance, load pattern, battery SOC to verify the success of the algorithm.
Step 10	Cost and impact analysis	To create a budget, analyzing feasibility and impact on the social and technological aspect.
Step 11	Analyzing future development scope	The project future development scope is open ended for induction charging, vehicle to grid solution etc.

Design description:

The system has been designed based on integration of Solar PV with a grid to supply load for Ev Vehicle charging, Battery Charging and microgrid power supply. The system component is divided mainly on 4 parts:

1. Solar to DC bus with MPPT controller
2. AC grid to DC bus through 3-phase rectifiers
3. DC bus to Battery system controlled by Battery Management system
4. DC bus to DC load (EV) and DC to AC load (microgrid)

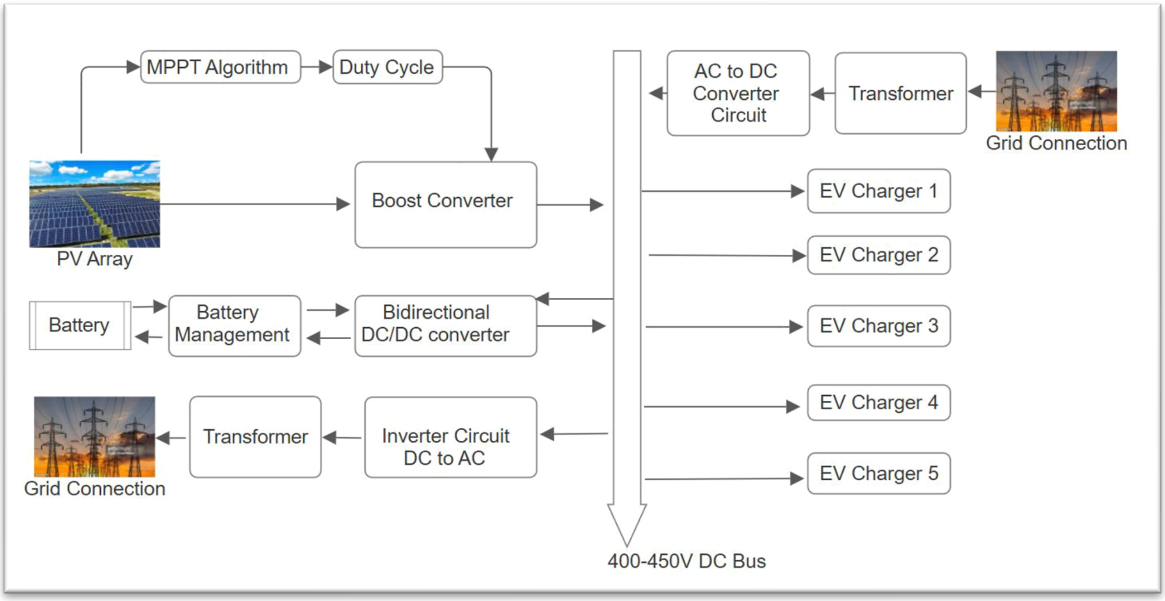

Figure 1: Design Planning

Load analysis:

- Type: DC load
- Number of Chargers: 5Capacity per Charger: 150 kW, 400V
- Total Load Capacity: **750 kW**
- Charging Standard: CCS (Combined Charging System) • Connector Type: CCS1
- Charging Time: Typically, 20–30 minutes for 20% to 80% State of Charge (SOC), depending on the vehicle's battery.
- Charger Type: Conductive (Plug-in)

Solar to DC bus with MPPT controller:

- **Solar PV System**
    - Planned Solar Load Capacity: 600 kW
    - Daytime Load Consideration: 450 kW over 10 hours, requiring 4500 kWh of energy.
    - Solar Capacity Calculation: If the solar peak period lasts 8 hours, then the required capacity is 4500 kWh / 8 h = 562 kW.
    - System Specifications: Output Voltage: 400V DC
    - Irradiance: 200–1000 W/m^2
    - DC bus voltage: 400 V (+/- 5%)

- Rated Power per Panel: 375W
- Operating Voltage (Vmp): 39.8V
- Open Circuit Voltage (Voc): 47.6V
- Operating Current (Imp): 9.4A
- Short Circuit Current (Isc): 9.96A
- Reference: Canadian Solar inc. CS3U-375MS-AG
- ***MPPT description:*** An MPPT for maximum power transfer has been implemented in boost converter. The Maximum Power Point (MPPT) is the point on the power-voltage curve where the solar produces the maximum power. The MPPT algorithm is designed to continuously track and adjust the operating point so that the system operates at this maximum power point under all conditions.
- ***MPPT algorithm:*** We have used Fuzzy Logic-based Maximum Power Point Tracking (Fuzzy MPPT) which is an advanced technique used to track the maximum power point of a solar panel. Unlike traditional methods such as Perturb and Observe (P&O) or Incremental Conductance (Inc.Cond), depend on mathematical formulas mostly, Fuzzy MPPT utilizes a fuzzy logic controller (FLC) that makes decisions based on linguistic variables. The algorithm has been given in a Simulink file and in the appendix. It has two inputs Δ V and Δ P. The real-time values of ΔV and ΔP are converted into fuzzy sets using predefined membership functions. If ΔV and ΔP is positive, then the power point is moving toward the maximum, and the control output should adjust accordingly to increase power. If ΔV and ΔP is negative, the system is moving away from the MPP, and the control output should change to steer the system back toward the MPPT. We have used 7-level fuzzy logic based on error threshold and controlled output by varying duty cycle.
- ***Boost converter:*** The reason of using boost converter is to reach a stable voltage despite conditions changed in solar panel due to irradiance, temperature, or shading effect. The boost converter can be controlled by MPPT to reach desired stable voltage output in DC bus. We have designed the boost converter components like inductor, capacitor and IGBT gate based on proven analytical equations.

- **AC grid to DC bus through 3-phase rectifier:**
    - The AC 3-phase voltage 208V,60Hz is connected to a DC bus through a 3-phase transformer of 208V AC to 400V AC and a 3-phase rectifier to convert the AC voltage to 400V DC voltage.

- *Rectifier:* A 3-phase rectifier generally employs six IGBTs, with two IGBTs for each phase—one for the positive half-cycle and one for the negative half-cycle of each phase. The pulse generator generates control signals for all six IGBTs, ensuring that their switching is synchronized. This synchronized switching guarantees that the IGBTs turn on and off in the correct sequence, enabling the proper functioning of the rectification process. Proper synchronization of the IGBT switching is crucial to prevent short circuits and ensure a smooth flow of current from the AC source to the DC load.

- **DC bus to Battery system controlled by Battery Management system:**
    - The DC 400 V is connected to a BESS for storing energy while additional solar power is available after feeding loads and storing the energy to supply during peak hours and lower solar availability. The battery system incorporates a buck-boost arrangement to charge and discharge the batteries based on the state of charge method. We have used Lithium-ion Batteries with 1415 Ah capacity and an initial state of charge of 50%.
    - ***BMS integration:*** A battery management system has been designed to ensure the battery charge or discharge based on its state of charge. The algorithm has been provided in a Simulink file and attached in the appendix:
    - ***Logic explanation:*** Disable boost if SoC < 0.2, disable buck if SoC > 0.8. If the value of the SOC is less than 20%. it would disable the boost function, and the battery would need to be charged. When the converter is in boost mode (i.e., boost_mode is true) and the current (I_bat) exceeds the current limit (I_lim), the duty cycle of the boost converter (duty_T1) is scaled down. The duty cycle (duty_T1) is multiplied by a factor of (I_lim / I_bat). This reduces the duty cycle to bring the current under the specified limit

- **DC bus to DC load (EV):**
    - ***EV Charger design:*** There are 05 EV Chargers, each designed for a 400 V and 375Ah battery. The initial SOC is considered 48.5%. A CCCV (Constant current constant voltage) Technology has been adopted. This technology is a two-stage technology widely used for ensuring the safety of battery charging. Initially the battery is supplied by a constant current and charge the battery without overheating. The battery voltage also rises gradually with current flow. Once the battery reaches a certain threshold voltage, it shifts to constant voltage mode to prevent overcharging and prolong battery life. In our project, we have set the constant current value to 280 A and the constant voltage 453 V. The EV charger is consists of a Boost converter, A PI controller, and Pulse generator to control the IGBT and display to present

the charging status. For simulation, we have kept option to change the load by a switch to demonstrate the load sharing from PV, and grid.

- ***Energy Management System:*** To control and monitor the load flow from different sources and distribute loads under different conditions, an EMS is applied based on logic and programmable control. A switch is used to change the number of DC loads in the system and to demonstrate how the loads are fed from PV and the grid based on logic. The microgrid activation and control are also implemented by EMS. The EMS logic has been focused on the logic, if Pload<Ppv, the loads will be fed by PV power only. If the power supply Pload>Ppv, the grid will come into the picture to supply power to the load. The microgrid will be activated when there is no DC (EV) load, and the battery State of charge is 80%. Light indicators are incorporated in the project to show status at different conditions and scenarios.
- ***DC BUS to Microgrid****:* The DC bus is also connected to a microgrid to supply local stations or nearby local loads to ensure maximum utilization of the system. The microgrid is controlled through the EMS logic. There is an interlocking switch system to cut off the other part of the system while the microgrid is fed. To convert the DC load to AC power required for a microgrid of 208 V 3-phase AC, we have used a 3-phase inverter with PWM generators.

5| Product Scope and Final Product Functionality

The scope of this project encompasses the design and simulation of a smart electric vehicle (EV) charging station integrated with a solar photovoltaic (PV) system, a battery energy storage system (BESS), and the utility electrical grid. The goal is to develop a reliable, sustainable, and intelligent charging infrastructure capable of supporting DC fast charging while optimizing energy flows from renewable and non-renewable sources.

The key components within the scope include:

- A 600kW solar PV array connected to a DC bus through a boost converter and fuzzy logic-based Maximum Power Point Tracking (MPPT) controller.
- A bidirectional DC-DC converter for battery integration, managed via a State of Charge (SoC)-based Battery Management System (BMS).
- A 400 V DC bus serving as the central voltage level for the charging station, with grid input via a step-up transformer and PWM-controlled rectifier.

- Integration of a microgrid through a three-phase inverter, allowing conversion back to 208 V AC for auxiliary or grid services.
- An Energy Management System (EMS) in progress, aimed at optimizing the use of solar, battery, and grid energy based on load demand and availability.
- A conceptual DC fast charging interface based on a Constant Current Constant Voltage (CCCV) protocol.

Simulation of the entire system was performed in MATLAB/Simulink, with detailed models for PV behaviour under variable irradiance, battery charging/discharging cycles, and power flow across different subsystems.

Functional Scope:

- Achieving and maintaining a stable 400 V DC output for reliable EV charging.
- Prioritized energy routing: solar → battery → grid.
- Dynamic power regulation using fuzzy logic MPPT.
- Control and protection of battery charging/discharging.
- Simulation-based validation of system performance under varying load and irradiance.

Limitations and Out-of-Scope Elements:

- Real-time hardware implementation and physical prototyping were not part of this scope.
- Industry-standard EV charger protocols (e.g., CHAdeMO, CCS) were not implemented due to restricted data availability.
- The EMS was partially implemented; advanced features like load forecasting or grid pricing algorithms remain future work.
- Vehicle-to-Grid (V2G) capability is identified as a potential future enhancement but was not included in the current design.
- Protection schemes such as surge protection, ground fault detection, and hardware-specific safety elements were beyond the project scope.

This defined scope ensured focused development of a simulation-ready EV charging infrastructure powered by renewable energy and guided by intelligent control algorithms.

6| Technical Specifications of the Final Product

The final product is a simulation-based model of a smart EV charging station integrated with solar PV, battery storage, and grid connectivity. The system was developed and validated in MATLAB/Simulink, incorporating realistic component behaviour and control logic to simulate a functional DC microgrid supporting fast EV charging.

The following are the major technical specifications of the final design:

Table 2: Core Electrical Specifications,

Component	**Specification**
Solar PV Array	200 parallel strings × 8 series modules = 600 kW total output
MPPT Controller	Fuzzy logic algorithm implemented in MATLAB Simulink
Battery Nominal Voltage	Compatible with 400 V DC bus, with bidirectional converter
Battery Bank	Rated to support peak loads with SOC-based charging/discharging
DC Bus Voltage	400V (stable, verified through simulation)
DC-DC Converter	Bidirectional converter for BESS with PID control
AC-DC Conversion	Transformer + PWM rectifier (208V AC → 400V DC)
DC-AC Inverter	3-phase inverter with voltage/current control logic (400V DC → 208V AC)
EV Charger Output	CCCV-based logic; fast charging simulations implemented
EMS	Partially implemented: logic for prioritizing solar > battery > grid
BMS	SOC-based PID control with over-current protection (simulated)
Simulation Environment	MATLAB/Simulink (including control logic and power flow)
Power Electronics and Control Systems	
DC-DC Converter	Boost converter for PV to DC bus, bidirectional buck/boost converter for battery integration.
MPPT Controller	Fuzzy logic-based MPPT algorithm with adaptive duty cycle scaling and dead-zone control around 400 V DC.
Rectifier	PWM-controlled 3-phase rectifier for converting 208 V AC grid supply to 400 V DC.
Inverter	3-phase voltage-source inverter connected to microgrid; implemented with sine wave generation through voltage-current logic control.
Transformer	Used for stepping up grid voltage prior to rectification, simplifying AC to DC conversion.

Battery Management System (BMS)	
SOC-based control logic	Charges or discharges based on SOC thresholds (e.g., disables boost mode if SOC < 0.2).
PID Controller	Used to manage charging current and voltage with anti-windup mechanism and current limiting.
Bidirectional control	Supports both charging (boost mode) and discharging (buck mode).

Table 3:Design Deviations and Justifications:

Change	Justification
SEPIC Converter → Boost Converter	SEPIC showed unstable voltage; boost provided better regulation.
DC Link eliminated	Transformer used for simplicity in grid-side connection.
Single inverter used in simulation	Real-world setup would require multiple parallel inverters for high current; not feasible in simulation.
V charging protocol simplified to CCCV	Lack of access to proprietary industrial data for standards like CHAdeMO/CCS.
EMS functionality limited	Focused on energy flow logic; advanced forecasting or pricing-based logic planned for future work.

Individual component ratings:

Table 4: PV Solar Pannel,

Module:	Canadian Solar Inc CS3U-375MS-AG
Parallel strings	200
Series-connected modules per string	8
Maximum Power (W)	375.3 W
Cells per module (Ncell)	72
Open circuit voltage Voc (V)	47.6
Short-circuit current Isc (A)	9.92A

Voltage at maximum power point Vmp (V)	39.8V
Current at maximum power point Imp (A)	9.43A
Temperature coefficient of Voc (%/deg.C)	-0.293
Temperature coefficient of Isc (%/deg.C)	0.035005

Table 5: Battery,

Battery type	Lithium-Ion
Nominal voltage (V)	400
Rated capacity (Ah)	1400
Initial state-of-charge (%)	50%
Battery response time (s)	0.1
Maximum capacity (Ah)	1400
Cut-off Voltage (V)	300
Fully charged voltage (V)	465.59
Nominal discharge current (A)	608.69
Internal resistance (Ohms)	0.00286
Capacity (Ah) at nominal voltage	1266.08

Table 6: Bidirectional DC-DC converter,

IGBT T1	
Resistance Ron (Ohms)	1e-3
Snubber resistance Rs (Ohms)	1e5
Snubber capacitance Cs (F)	inf
IGBT T1	
Resistance Ron (Ohms)	1e-3
Snubber resistance Rs (Ohms)	1e5
Snubber capacitance Cs (F)	inf
Off state the IGBT model impedance.	infinite

C0 Capacitance (F)	0.01
Cin Capacitance (F)	0.015
Resistance (Ohms)	0.00005
Inductor	
Inductance (H)	4e-4

Table 7: MPPT Boost Converter;

IGBT	
Resistance Ron (Ohms)	0.001
Inductance Lon (H)	0
Forward voltage Vf (V)	0
Initial current Ic (A)	0
Snubber resistance Rs (Ohms)	1e5
Snubber capacitance Cs (F)	inf
Off state the IGBT model impedance.	infinite
Capacitor	
C1 Capacitance (F)	0.00018
C2 Capacitance (F)	0.04
Inductor	
Inductance (H)	1.8e-05
Diode	
Resistance Ron (Ohms)	0.001
Inductance Lon (H)	0
Forward voltage Vf (V)	0.8
Snubber resistance Rs (Ohms)	500
Snubber capacitance Cs (F)	250e-9
Resistor	
Resistance (Ohms)	0.28

Table 8: AC-DC Converter:

Transformer	
Winding 1 connection (ABC terminals)	Yg
Winding 2 connection (abc terminals)	Yg
Transformer type	3-phase
Nominal power and frequency [Pn(VA) , fn(Hz)]	[1000e3, 60]
Winding 1 parameters [V1 Ph-Ph(Vrms) , R1(pu) , L1(pu)]	[296, 0.002, 0.08]
Winding 2 parameters [V2 Ph-Ph(Vrms) , R2(pu) , L2(pu)]	[208, 0.002, 0.08]
Magnetization resistance Rm (pu)	500
Magnetization inductance Lm (pu)	500
Thyristor	
Resistance Ron (Ohms)	0.001
Inductance Lon (H)	0
Forward voltage Vf (V)	0.8
Initial current Ic (A)	0
Snubber resistance Rs (Ohms)	500
Snubber capacitance Cs (F)	250e-9
Pulse Generator	
Pulse type	Time based
Amplitude	1
Period (Sec)	0.016
Pulse Width (% of period)	33.33
Pulse delay (Sec)	0.013333
Resistor	
Resistance (ohms)	10

Table 9: DC-AC Inverter:

IGBT	

Internal resistance Ron (Ohms)	1e-3
Snubber resistance Rs (Ohms)	1e5
Snubber capacitance Cs (F)	inf
Diode	
Resistance Ron (Ohms)	0.001
Inductance Lon (H)	0
Forward voltage Vf (V)	0.8
Initial current Ic (A)	0
RLC Filter	
Inductance (H)	5.1e-3
Resistance R (Ohms)	5.34
Capacitance C (F)	9.94e-6

7|Measuring Success and Validation Test Results

TEST and Validation: Solar output:

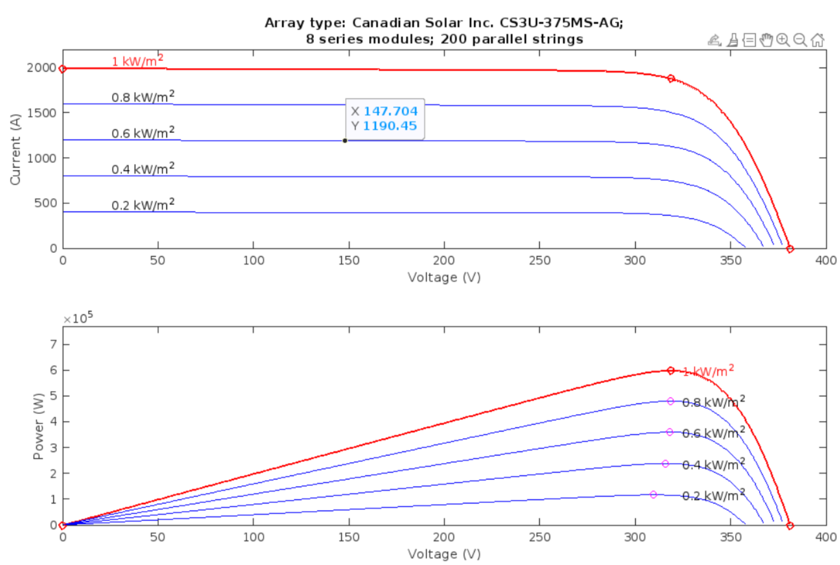

Figure 2: Current and power at different irradiance

In this figure 2, the power and current output at different irradiance have been shown at the maximum power point (MPP). We can observe that 1000 watts/m2, the power output is approximately 600 kW, and current is 2000A. With the change of irradiance, the value will be adjusted, and maximum power will be extracted based on the MPPT method.

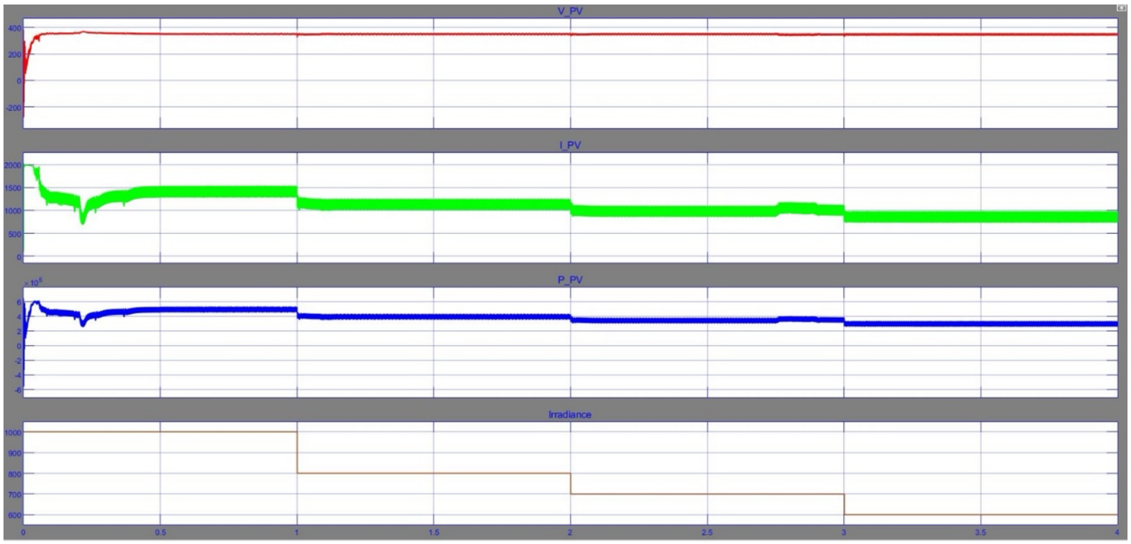

Figure 3: Power and Current Output at Different Irradiance

The figure(3) shows, the output voltage of PV, the output current of PV, the Current in diode, and a specific irradiance of 1000 watt/m2 and temperature of 25 deg-Celsius. From the figure we can see the output voltage of PV is 380 V which will be boosted to 400V after Boost converter.

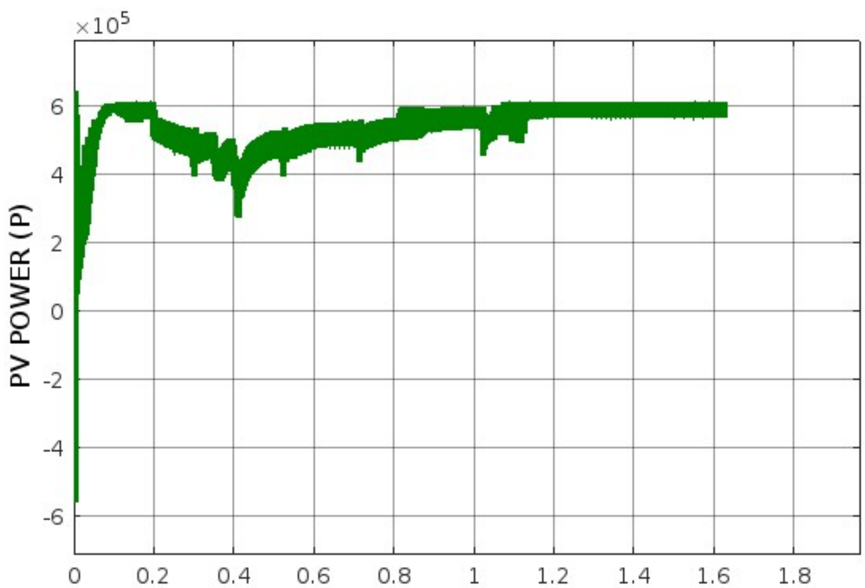

Figure 4: PV Output w.r.t 1000 Watt/m2

The figure 4 shows the PV power output at 1000 Watt/m2 irradiance. From the figure, we observe the power output is around 580 kW, which was desired based on the proposal.

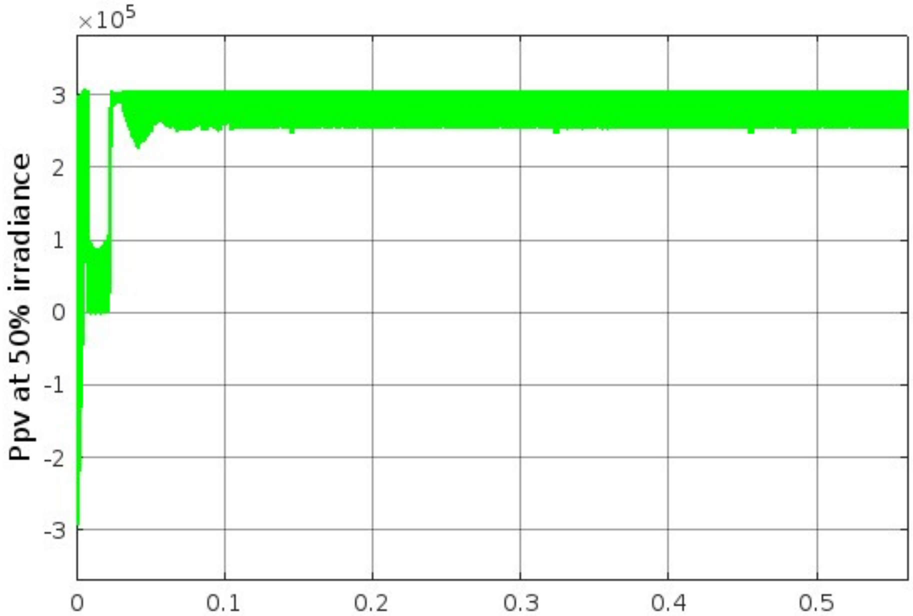

Figure 5: Power Output w.r.t 50% irradiance

This figure 5, demonstrates the power output reduction due to less irradiance (50%), The output consequently became around 300 kW, which indicates the PV output relation with irradiance.

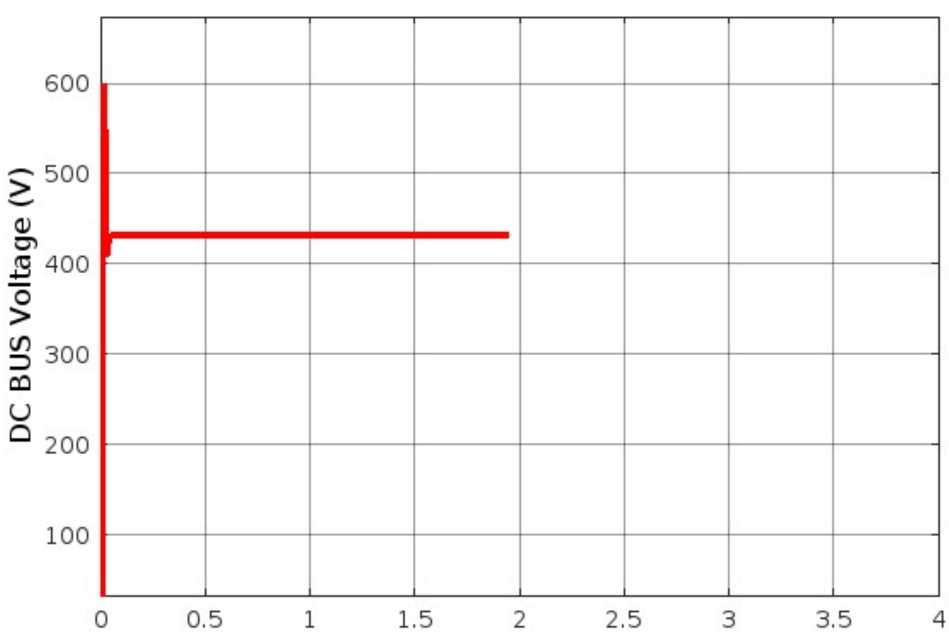

Figure 6: DC Bus Voltage

In this figure 6, we have achieved the DC BUS voltage of the system after the boost converter as 420 V, which is within the limit of the fixed DC bus voltage for charging the EV vehicles. The EV charging station is connected to this bus.to keep the bus voltage constant to 400V +-5% is a validation of the system throughout the simulation.

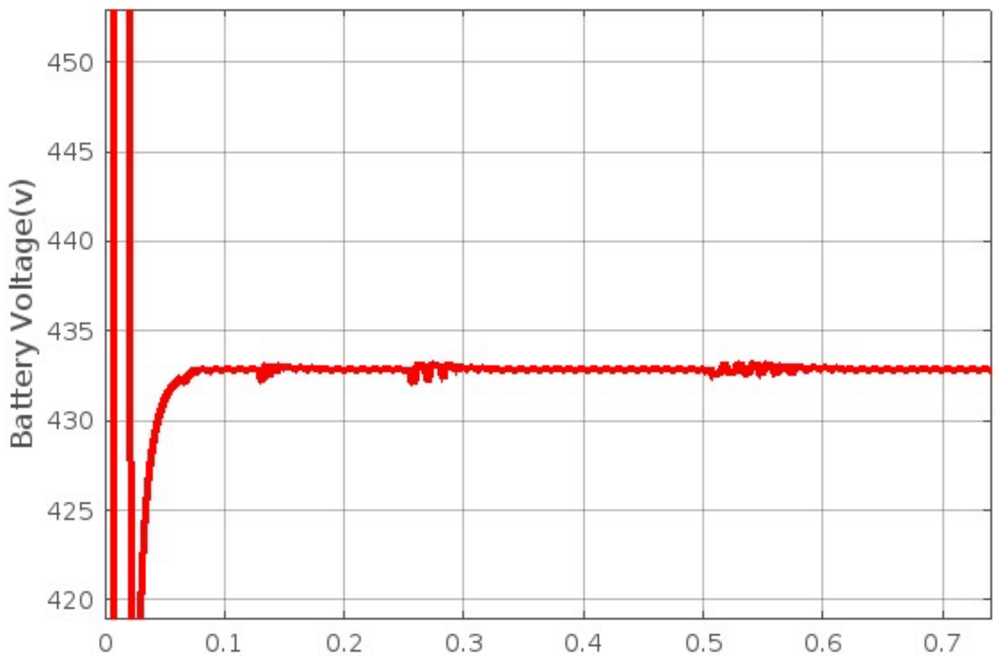

Figure 7: Battery Output Voltage

The Battery Output voltage is designed such that it maintains stability with DC bus voltage, as it will be synchronized with the DC bus and charge and discharge to that node. From the figure, we can see the DC battery voltage is approximately to a similar value of 420 V in the Dc bus.

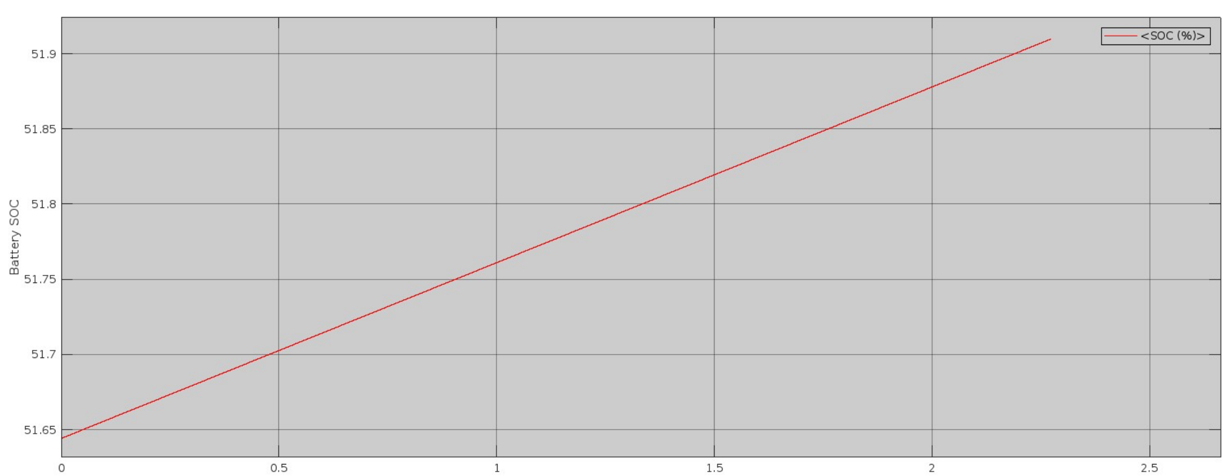

Figure 8: Battery SOC (State-of-charge) indication

The figure 8 indicates the SOC rising from the initial set value of 50% to up, which indicates the battery is in charging status.

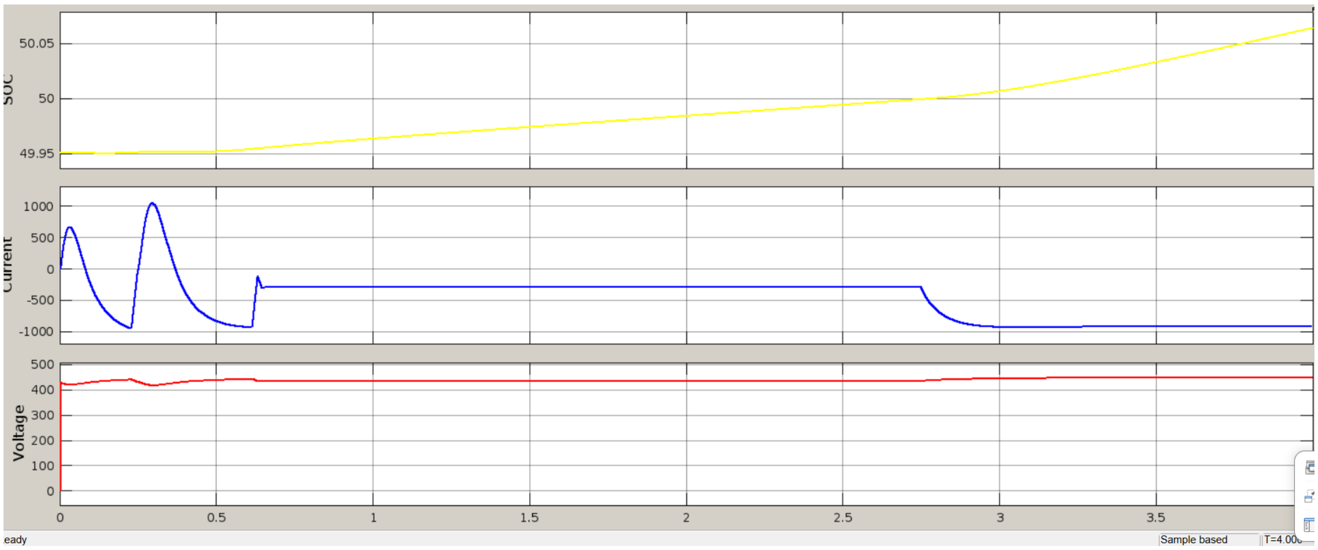

Figure 9: SOC w.r.t Current and Voltage

The figure 9 indicates the EV charger battery following the Constant current constant voltage method. The current has been set up to be charged at a maximum rate of 280 A (fixed based on the controller) to overcome over charging process. The voltage was kept constant at 400-450 volts to prolong battery life.

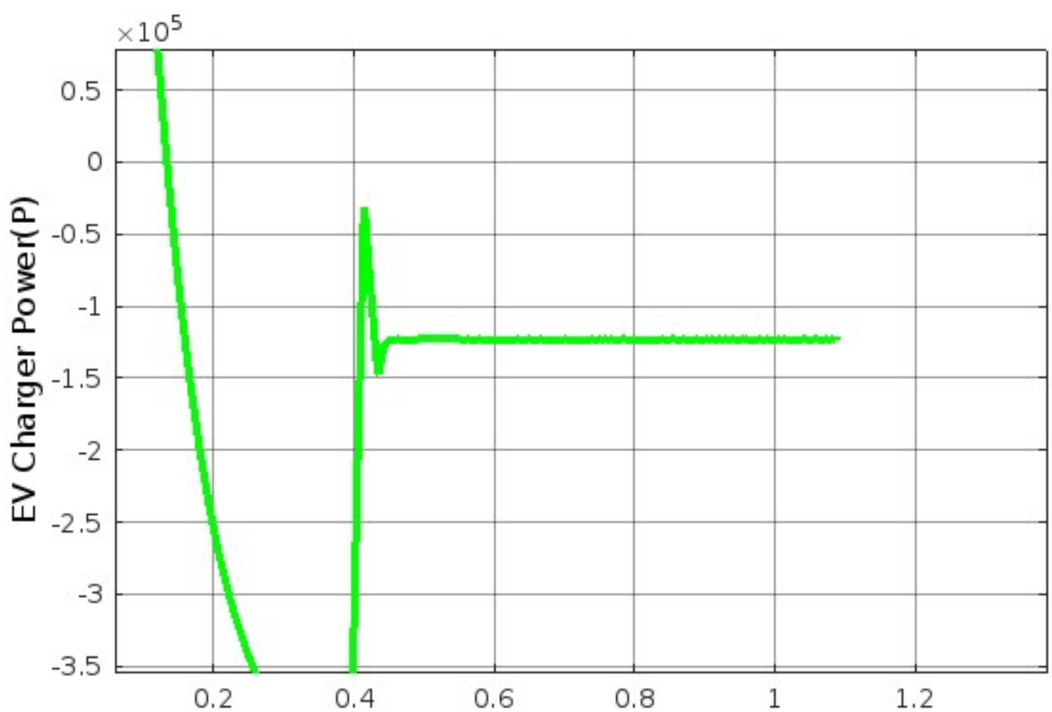

Figure 10: Single EV Charger Power Output

The figure shows the EV charger power for a single charger, which indicates charging at around 120-130kW.

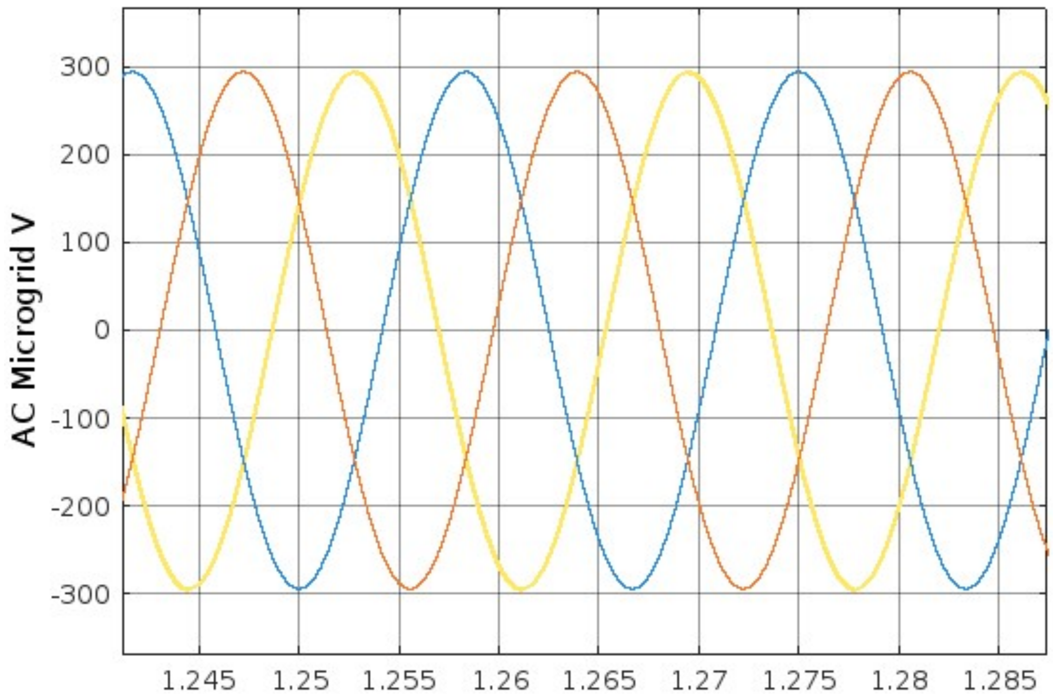

Figure 11: AC Grid Voltage (3-phase, 208V)

The figure 11, indicates the AC Grid voltage in the microgrid side. The required Voltage is 3-phase 208 V is achieved after the 3-phase inverter circuit, that converts the DC voltage to AC voltage.

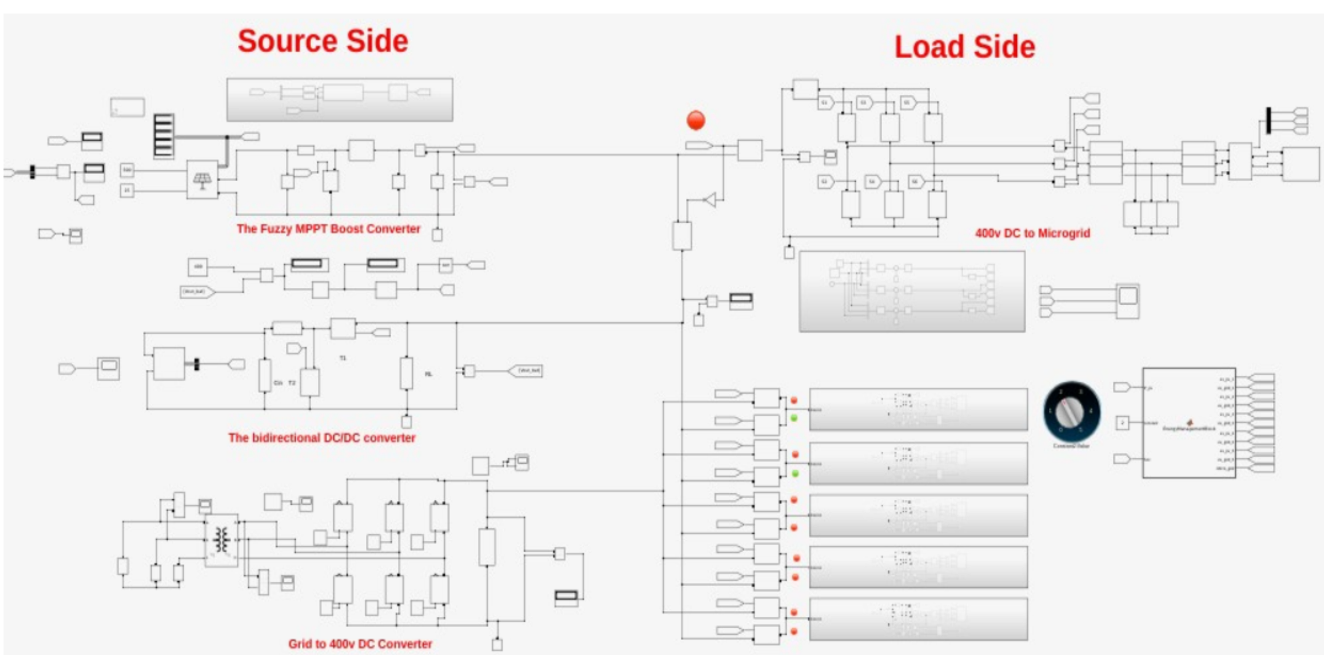

Figure 12: MATLAB/Simulink Schematic Diagram (Source Side and Load Side)

In this figure, the implementation of EMS has been shown. First, we selected two chargers working together through the selector switch, and during charging of only two vehicles, the DC load is lower than PV generation, and thus two green indicators show the power has been received from solar PV only. The red mark on the microgrid side shows microgrid is out of the system under these conditions.

8| List of Tools, Materials, Supplies Costs, etc

Table 10: Cost Analysis

Solar plant cost					
Material	Unit cost ($)	Unit	Quantity	Total cost	Remarks/reference
Module	$0.38	W	700,000 W	266000	SEIA (2024). Solar Market Insight Report Q2
Racking (Ground Mount)	$22	m²	3,000m²	66,000	NREL (2024). Solar Cost Benchmarks (NREL/TP-6A20-89710)
DC Cables (4 AWG)	$1.50	m	2,000m	3,000	Supplier Quote (ACME Solar, 2024)
Electrical BOS	$ 27	sq. m	3000	81000	RSMeans (2022), NREL (2022)
Installation rental equipment	$3.85	sq. m	3000	11550	NREL (2022)
Installation labour	1.16	sq. m	3000	3480	NREL (2022)
Subtotal				431,030	
EPC Overhead	13%			56033.9	NREL (2022)
Total				487063.9	
Battery Energy Storage System					
	Unit cost ($)	Unit	Quantity	Total cost	Remarks/reference
Module (Battery cost)	$195	kWh	550	107,250	NREL (2024). Battery Storage Cost Report (NREL/TP-6A20-90412)
Inverter /converter price	97.5	kW	200	19500	NREL 2023

BMS integration	15	kW	200	3000	NREL 2023
Battery Enclosure	$8,000	nos	5	40,000	UL 1973 (2022). Standard for BESS
Total				169,750	
Inverter/converter/Transformer	Unit cost ($)	Unit	Quantity	Total cost	Remarks/reference
DC Converter	45,000	nos	2	90,000	Supplier
Circuit Breakers	1,200	nos	10	12,000	ABB (2023). S800PV Datasheet
Inverter/rectifier	60,000	nos	2	120,000	Supplier
Transformer	40,000	nos	2	80,000	Supplier
Relays	500	nos	6	3,000	SEL Technical Manual
Total cost				305,000	
EV Chargers (5×150 kW)					
Material	Unit cost ($)	Unit	Quantity	Total cost	Remarks/reference
DC Fast Chargers	$38,000	nos	5	$190,000	U.S. DOE (2024). Alternative Fuel Station Costs
Installation Labor	$8,000	nos	5	$40,000	Alberta Electrical Safety Code (2024)
Total				230,000	
Land Cost:(lease over a period of 50 years)	The cost of land is not considered a capital investment in this project. Instead, the plan is to lease the land for a period of 50 years, as most of it falls under federal government jurisdiction and is protected. The expenses associated with the lease will be covered by the revenue generated from selling or operating the charging facility				
Cabling cost					
HV Cables (Grid)	$15	m	500m	7,500	ATCO (2024). Technical Specifications
LV Cables	$8	m	300	2400	CSA C22.1-24 (2024 Electrical Code)
PV Array Cabling	$1.50	m	2,500	3,750	Supplier Quote (ACME Solar)

Conduits & Raceways	$6	m	800	4,800	Alberta Safety Codes Act
Grounding System	$2,000	nos	1	2,000	IEEE Standards
Total				20,450	
Total materials cost for the project				1212263.9	
Additional cost (licensing, land acquisition, tax, project cost, finance cost, grid integration) approximately 20%				242,452.78	
Total approximate cost for the project				1.45 million dollars	

- **Solar PV System Costs**
  - Panel price drop from 0.40/W to 0.38/W (SEIA 2024 vs. NREL 2022 data)
  - Structural BOS costs reduced from 25/m2 to 22/m^2 (NREL 2024 benchmarks)
- **Battery Storage**
  - Batteries: reduced from 215/kWh to 195/kWh
- **EV Charger**
  i. Added installation labor cost.
- **Added overall protection materials Cost**
- **Cabling & Distribution**

  Newly Added:

  i. HV and LV cables
  ii. Grounding system for safety compliance
  iii. Conduits for outdoor durability

Total project cost is 1.4547 million dollars which is 23.4465% less than the price of design roadmap.

Supplier List

Table 11: Solar PV System Suppliers:

Material	**Supplier 1 (Local/CA)**	**Supplier 2 (Global)**	**Notes**
Solar Panels (700kW)	Canadian Solar	SunPower	Silfab (BC) also available

Material	Supplier 1 (Local/CA)	Supplier 2 (Global)	Notes
Racking (Ground Mount)	Terrasmart Canada	IronRidge	Eclipsall (ON) for custom solutions
DC Cables (4 AWG)	Eecol Electric (Calgary)	Prysmian Group	Westburne (Calgary) alternative
Electrical BOS	Schneider Electric	SMA America	Includes combiners/disconnects

Table 12: Battery Storage Suppliers:

Material	Supplier 1 (Local/CA)	Supplier 2 (Global)	Notes
Battery Modules (550kWh)	Eguana Tech	Tesla Powerwall	LG Chem also available
Inverters (200kW)	Fronius Canada	SolarEdge	OutBack Power for off-grid
Battery Enclosures	Hammond Mfg	Dynapower	NEMA 4X rated

Table 13: EV Charging Station Suppliers:

Material	Supplier 1 (Local/CA)	Supplier 2 (Global)	Notes
150kW Fast Chargers	FLO	ChargePoint	Tritium also an option
Installation Labor	Amped EV Solutions (Calgary)	Black & Veatch	Local electricians preferred

Table 14: Cabling & Distribution Suppliers:

Material	Supplier 1 (Local/CA)	Supplier 2 (Global)	Notes
HV Cables (Grid)	Southwire Canada	Nexans	Eecol stocks these
Conduits	Thomas & Betts	Atkore	PVC-coated for durability
Grounding System	ERICO Canada	ABB	UL/CSA compliant

9| Reflection on Ethics and Equity in your Capstone Journey

Ethics and equity played a fundamental role in shaping the decisions made throughout our project. Given that our system integrates solar energy with EV charging and battery storage, ethical considerations included ensuring system safety, sustainability, and compliance with engineering standards.

From an equity perspective, we focused on making renewable energy more accessible and ensuring fair distribution of power across different loads. The synchronization algorithm developed for our system prevents any single component—EV chargers, batteries, or grid loads—from being unfairly prioritized. This ensures all power demands are met efficiently while maintaining system stability.

In terms of safety, we implemented protective mechanisms such as overvoltage and overcurrent protection for the inverter and rectifier. The Battery Management System (BMS) was designed to prevent hazardous conditions such as overcharging, deep discharging, and thermal runaway. These considerations align with industry standards such as IEEE 1547 for interconnecting distributed energy resources and IEC 62133 for battery safety.

Environmental ethics also played a role in our decision-making. We prioritized energy-efficient power electronics and designed the system to optimize solar power utilization, reducing dependency on non-renewable energy sources. By ensuring our design aligns with ethical and equitable engineering principles, we have created a solution that is both technically effective and socially responsible.

10| Reflection on Impact of Engineering on Society/Environment

The adoption of solar energy and EVs is rapidly transforming modern energy infrastructure. Our project directly contributes to this transition by integrating solar-based charging with grid synchronization, reducing reliance on fossil fuels and promoting sustainability. The following table outlines the key societal and environmental impacts of our project:

Table 15: **Impact of Renewable Energy Integration and EV Adoption**

Impact Area	Description
Energy Accessibility	Solar integration ensures reliable power in areas with unstable grid supply, promoting decentralized energy solutions.
Environmental Impact	EVs and solar reduce greenhouse gas emissions and air pollution compared to conventional fossil fuel-based energy systems.
Economic Benefits	Widespread EV adoption lowers transportation costs, while solar power decreases electricity expenses over time.
Infrastructure Development	More EV charging stations and solar farms are needed to support adoption, requiring grid modernization and smart energy management.
Consumer Adoption	Awareness programs and government incentives (e.g., subsidies, tax credits) encourage users to transition to EVs and renewable energy.

By designing a system that optimizes renewable energy use for EV charging, our project contributes to sustainability efforts while addressing key challenges in energy management.

11| Reflection on Teams' Approach to Project Management

To effectively manage the project, our team adopted a structured approach that balanced workload distribution, milestone tracking, and problem-solving strategies.

Table 16: Project Management Approach

Project Management Aspect	Details
Workload Distribution	Tasks were assigned based on expertise: software simulation (MATLAB/Simulink) was handled by members with strong coding skills, while hardware design was led by those familiar with power electronics.
Key Milestones Achieved	1. Load assessment and system modelling (PV, battery, grid loads). 2. Component specification and selection (MPPT, inverter, rectifier). 3. Simulation of power flow and charging optimization. 4. Implementation and validation of battery management and energy distribution.

Project Steps	1. Research and requirement gathering → 2. System design → 3. Simulation and validation → 4. Testing and optimization.
Challenges and Solutions	Faced issues with component compatibility and system integration, which were resolved through iterative simulations and expert consultations.

By organizing tasks efficiently and setting clear milestones, we ensured steady progress while addressing technical challenges as they arose.

12| Reflection Teams' Approach to Economics and Cost Control

Economic feasibility was a critical factor in our project. Given the high costs associated with renewable energy systems and EV charging infrastructure, our team focused on cost-effective strategies without compromising efficiency.

- Budget Planning: We conducted a market analysis to determine the most cost-effective solar panels, batteries, and power converters.
- Cost Reduction Strategies: Instead of using pre-built commercial inverters and rectifiers, we designed a customized power conversion system that met our specific needs, reducing overall expenses.
- Energy Optimization: The MPPT (Maximum Power Point Tracking) algorithm ensures maximum power extraction from the solar panels, reducing the need for additional energy generation.
- Battery Life Optimization: The BMS prevents excessive discharge cycles, extending battery life and lowering replacement costs.

Additionally, we explored different economic models for EV charging infrastructure, including dynamic pricing based on grid demand and renewable energy availability. By prioritizing long-term sustainability, our design remains financially viable while promoting widespread adoption of clean energy.

13| Reflection on Team Members' Approach to Life-long Learning

Throughout this project, our team had to continuously acquire new knowledge and refine our technical skills to address emerging challenges. The interdisciplinary nature of our design required expertise in power electronics, renewable energy, energy management algorithms, and system modelling. Some of the key areas where we expanded our knowledge include:

- Power Electronics & Energy Systems: We deepened our understanding of inverter topologies, DC/DC converters, and energy flow optimization for bidirectional power transfer.

- Simulation and Software Skills: Extensive use of MATLAB/Simulink for modeling MPPT, BMS, and power flow simulation helped us refine our system design before physical implementation.
- Control Systems & Optimization: Learning and implementing different MPPT techniques (Perturb & Observe, Incremental Conductance, and Fuzzy Logic) allowed us to select the most stable and efficient approach.
- Industry Standards & Compliance: We reviewed key standards such as IEEE 2030.5 for smart grid interoperability and SAE J1772 for EV charging to ensure our system meets industry guidelines.

By leveraging online resources, research papers, and expert consultations, we adapted to new challenges and gained valuable skills applicable to future engineering projects. This project has reinforced the importance of continuous learning, as advancements in renewable energy and smart grid technologies will continue to shape the future of sustainable engineering.

14| Appendix

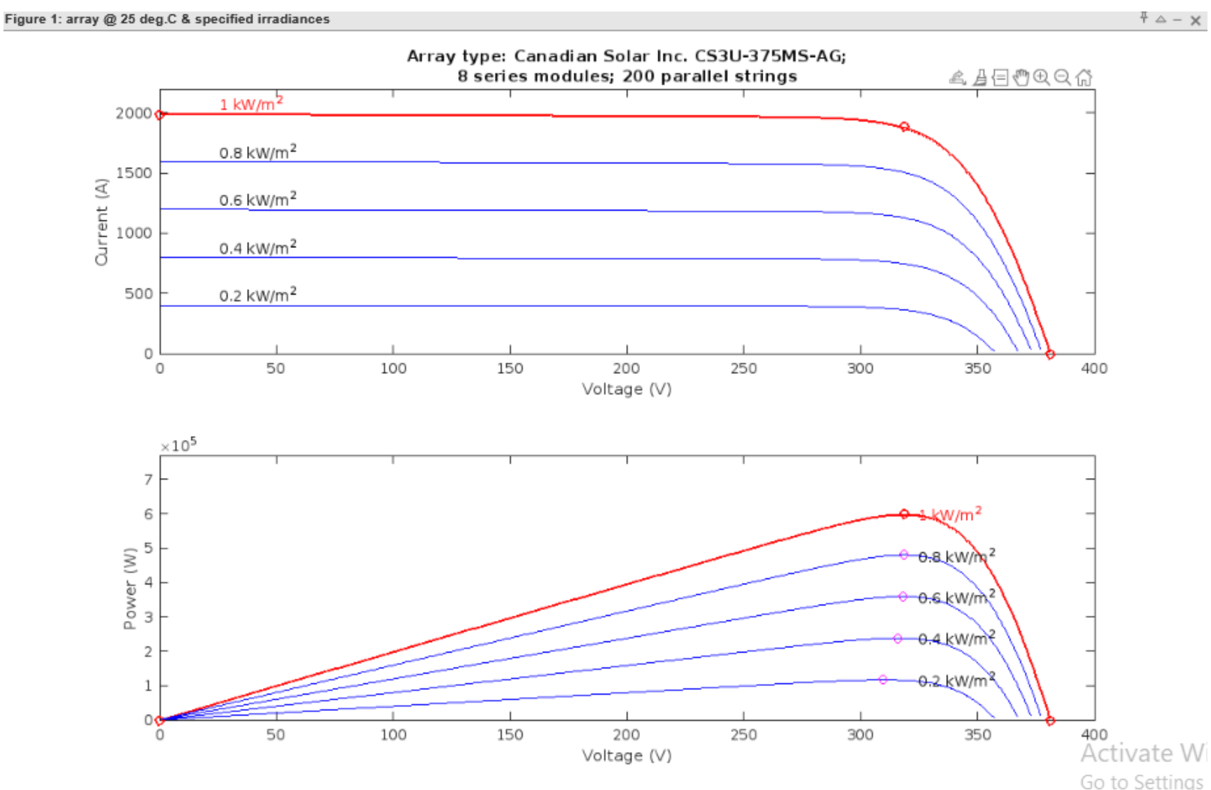

Figure 13: I-V curve for different solar irradiance

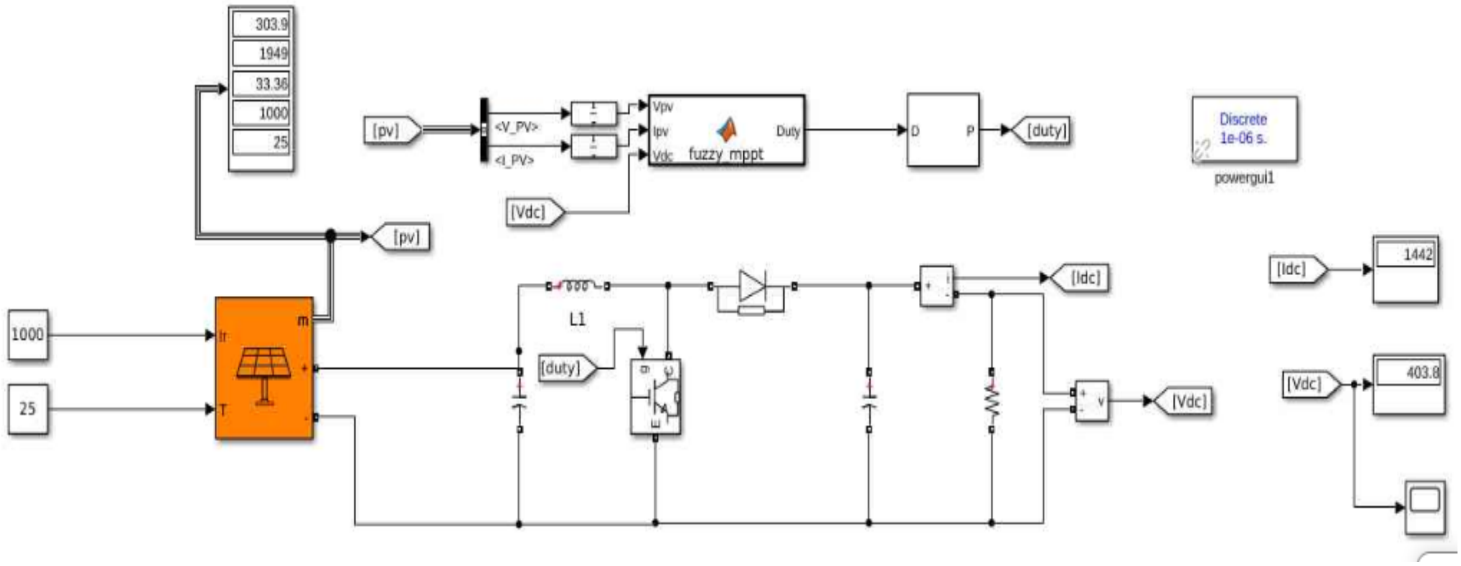

Figure 14: Diagram for MPPT simulation with solar modules outcome

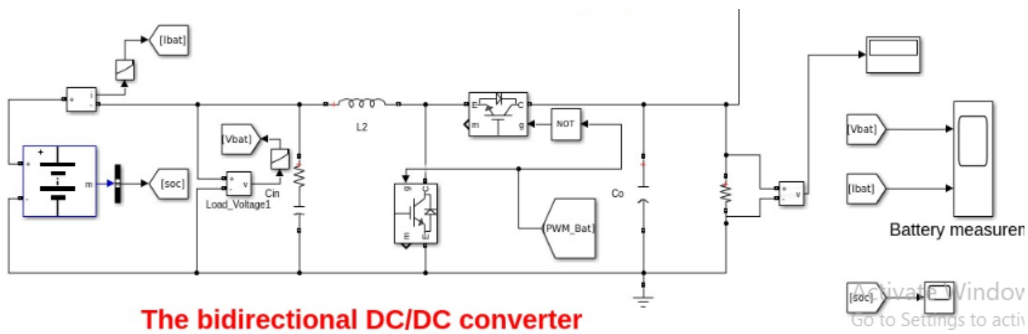

Figure 15: DC to DC Bidirectional Converter

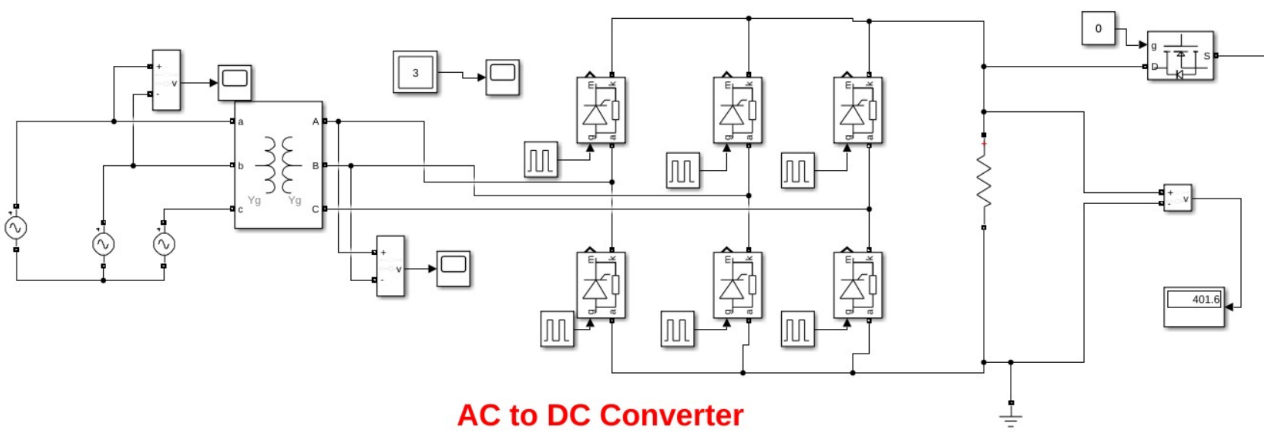

Figure 16: AC to DC Converter

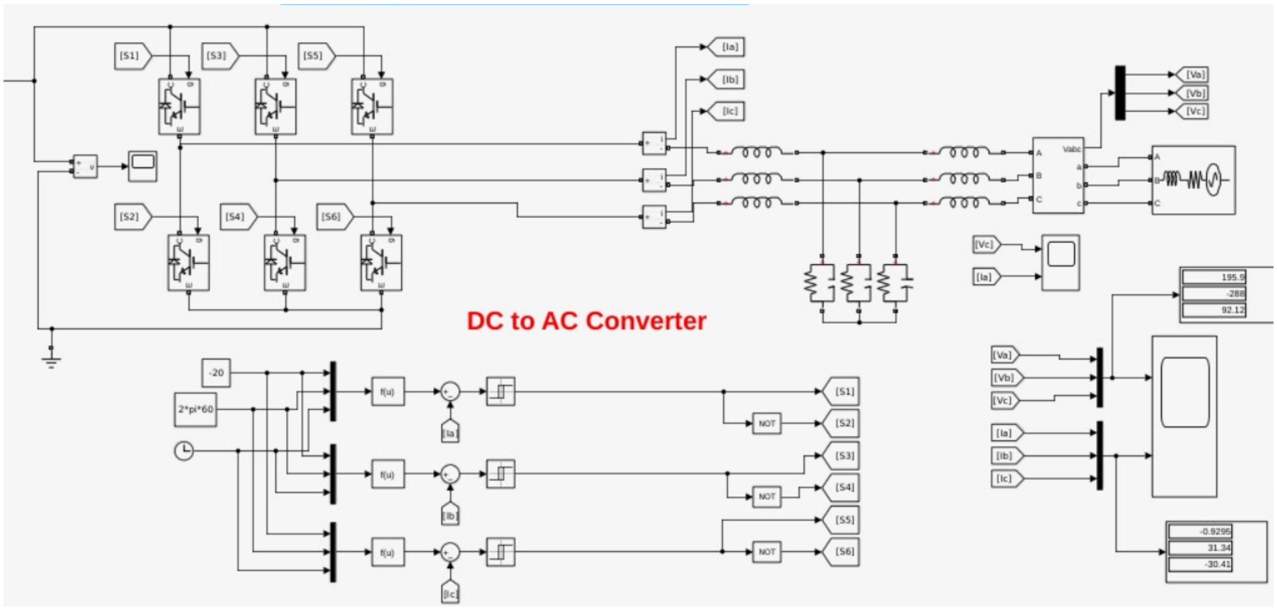

Figure 17: DC to AC Converter

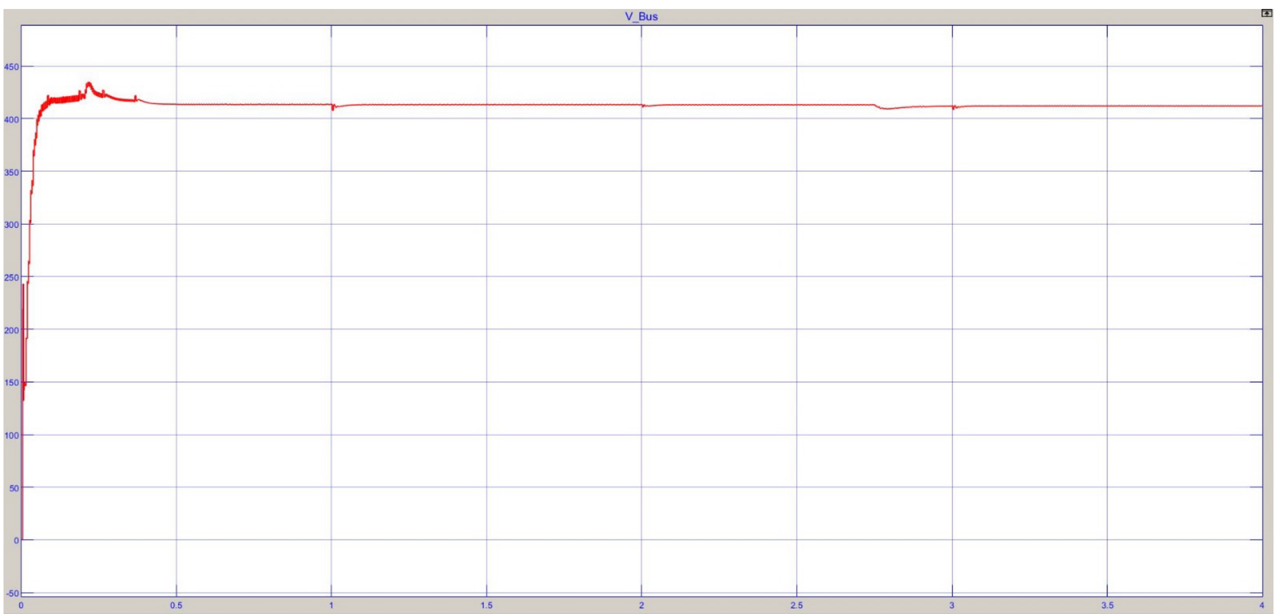

Figure 18: DC BUS Voltage

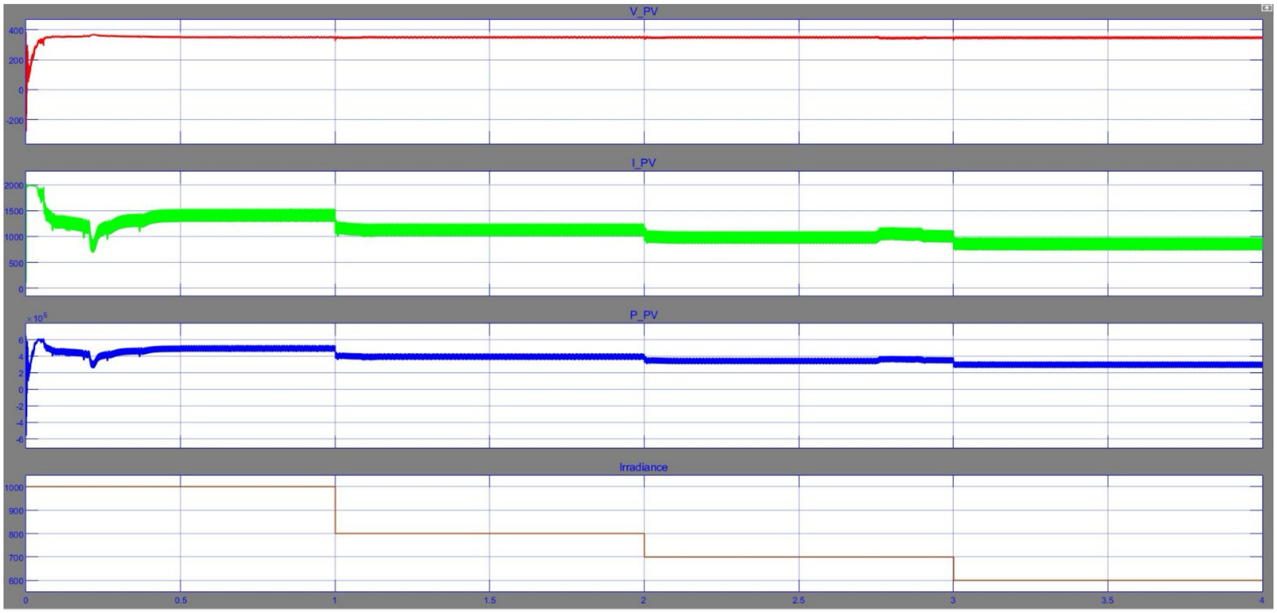

Figure 19: PV Output Voltage and Current w.r.t Irradiance

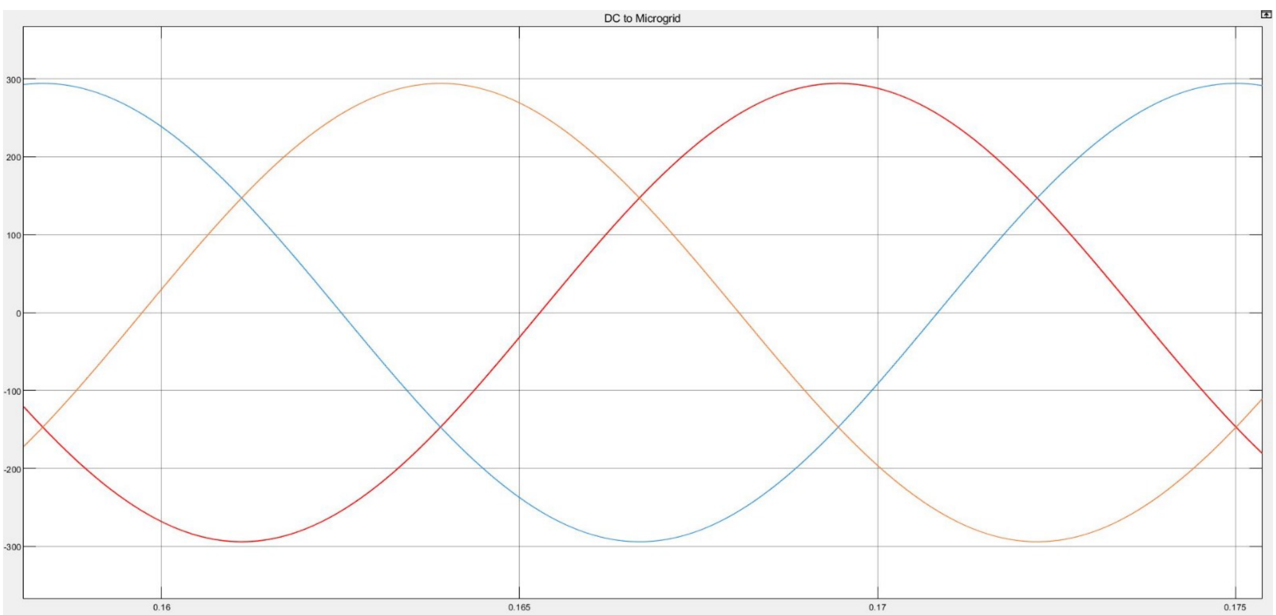

Figure 20: Microgrid Voltage

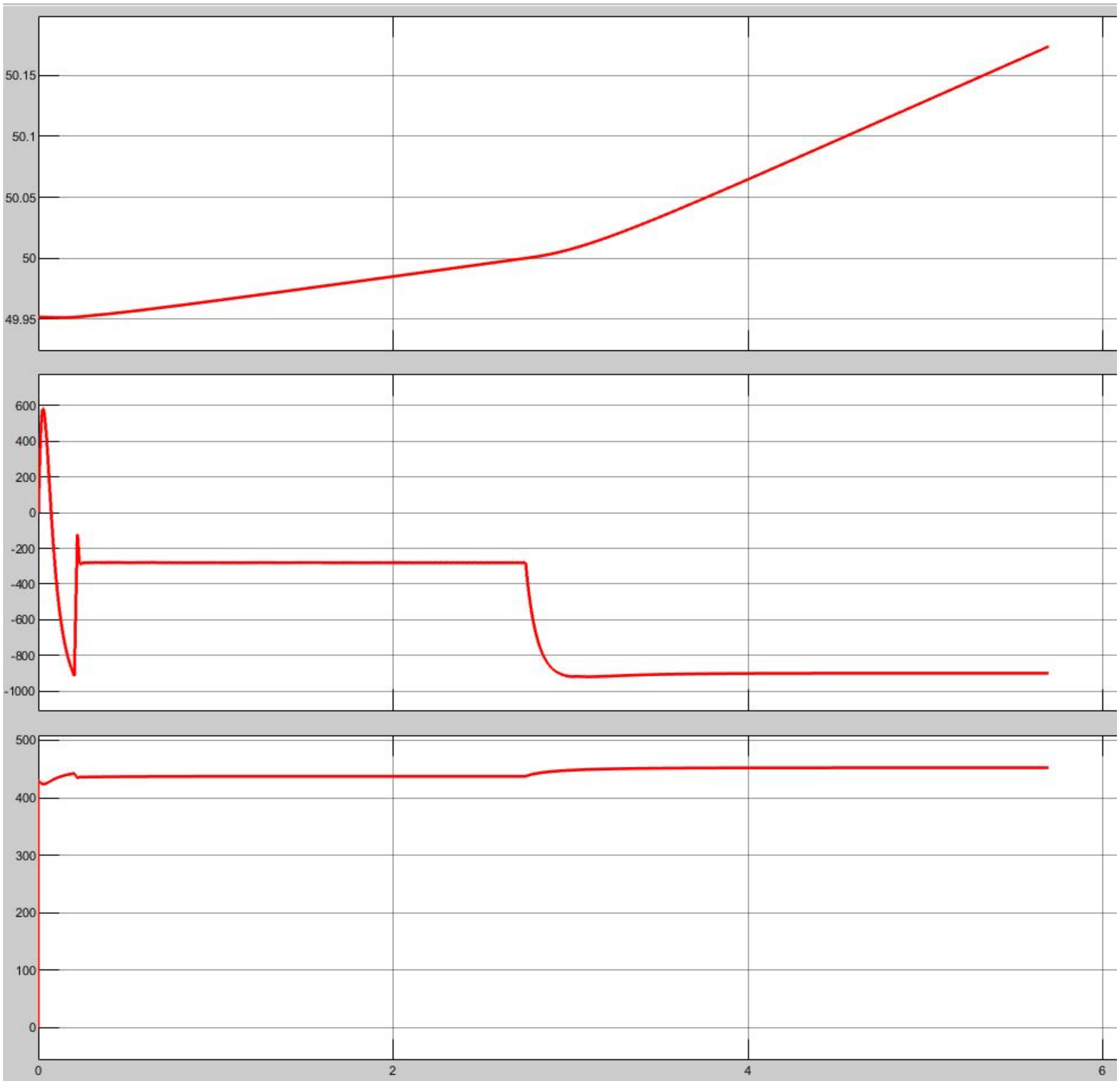

Figure 21: SOC of Battery

MPPT Algorithme:

```
function Duty = fuzzy_mppt(Vpv, Ipv, Vdc)
   % Persistent Variables to Store Previous Values
   persistent P_prev  E_prev Duty_prev;

   % Initialize Persistent Variables on First Run
   if isempty(P_prev)
      P_prev = 0;        % Previous Power
      E_prev = 0;        % Previous Error
```

```
    Duty_prev = 0.204;  % Initial Duty Cycle (20.4%)
end

% Calculate Instantaneous PV Power
Ppv = Vpv * Ipv;

% Compute Error (Difference in PV Power)
E = Ppv - P_prev;

% Compute Change in Error (Rate of Change of Power)
dE = E - E_prev;

% Define 7-Level Fuzzy Membership Functions for E and dE
NB = -3; NM = -2; NS = -1;
Z = 0;
PS = 1; PM = 2; PB = 3;

% Define error thresholds for improved accuracy
err_thresholds = [-20, -10, -5, 0, 5, 10, 20];  % Fuzzy boundaries
d_err_thresholds = [-15, -8, -4, 0, 4, 8, 15];  % Change in power boundaries

% Assign fuzzy levels based on error
if E < err_thresholds(1)
    E_fuzzy = NB;
elseif E < err_thresholds(2)
    E_fuzzy = NM;
elseif E < err_thresholds(3)
    E_fuzzy = NS;
elseif E < err_thresholds(4)
    E_fuzzy = Z;
elseif E < err_thresholds(5)
    E_fuzzy = PS;
elseif E < err_thresholds(6)
    E_fuzzy = PM;
else
    E_fuzzy = PB;
end

% Assign fuzzy levels based on change in error
if dE < d_err_thresholds(1)
    dE_fuzzy = NB;
elseif dE < d_err_thresholds(2)
    dE_fuzzy = NM;
elseif dE < d_err_thresholds(3)
    dE_fuzzy = NS;
elseif dE < d_err_thresholds(4)
    dE_fuzzy = Z;
elseif dE < d_err_thresholds(5)
```

```
    dE_fuzzy = PS;
elseif dE < d_err_thresholds(6)
    dE_fuzzy = PM;
else
    dE_fuzzy = PB;
end

% **Fuzzy Inference System (7x7 Rule Base)**
% Rule base: Mapping E_fuzzy and dE_fuzzy to dD
fuzzy_rule_table = [  NB, NB, NM, NS, Z,  PS, PM;
                      NB, NB, NM, NS, Z,  PS, PM;
                      NM, NM, NS, Z,  PS, PM, PB;
                      NS, NS, Z,  Z,  Z,  PS, PS;
                      Z,  Z,  PS, Z,  Z,  NS, NS;
                      PS, PM, PB, PS, Z,  NM, NM;
                      PM, PB, PB, PM, Z,  NB, NB  ];

% Get dD from fuzzy rule table
dD = fuzzy_rule_table(E_fuzzy + 4, dE_fuzzy + 4); % Offset by 4 to index correctly

% **Adaptive Scaling Factor for Duty Cycle Change**
base_scaling = 0.0004;  % Smaller step changes for ultra-fine control

% Adaptive Scaling: Reduce when power changes are high
if abs(dE) > 10
    scaling_factor = base_scaling * 0.5;  % Reduce step size when fluctuations are high
else
    scaling_factor = base_scaling;  % Normal operation
end

dD_actual = dD * scaling_factor;

% **Boost Converter Output Voltage Stability Control**
Vdc_ref = 405;  % Target Boost Output Voltage
Vdc_tolerance = 1;  % Allowable voltage range (strict but controlled)

% **Create a "Dead Zone" Around 400V**
% This prevents unnecessary tiny duty cycle changes due to noise
if abs(Vdc - Vdc_ref) < 1
    dD_actual = 0;  % Hold duty cycle constant in the "safe zone"
elseif Vdc > Vdc_ref + Vdc_tolerance   % If Vdc is too high, slightly decrease duty cycle
    dD_actual = dD_actual - 0.00005;  % Very fine correction
elseif Vdc < Vdc_ref - Vdc_tolerance   % If Vdc is too low, slightly increase duty cycle
    dD_actual = dD_actual + 0.00005;  % Very fine correction
end

% **Update Duty Cycle with Clamping**
Duty = Duty_prev + dD_actual;
```

```
    % **Clamp Duty Cycle Between Safe Limits (Preventing Instability)**
    if Duty > 0.85   % Reduce max limit to prevent excessive boost voltage
        Duty = 0.85;
    elseif Duty < 0.15   % Ensure enough switching action
        Duty = 0.15;
    end

    % **Store Updated Values for Next Iteration**
    P_prev = Ppv;
    E_prev = E;
    Duty_prev = Duty; % Store Updated Duty Cycle
End
```

Algorithm for EMS:

```
function [ev_pv_1, ev_grid_1, ev_pv_2, ev_grid_2, ev_pv_3, ev_grid_3, ev_pv_4,
ev_grid_4, ev_pv_5, ev_grid_5, Micro_grid] = EnergyManagementBlock(P_pv,
constant, Soc)
% EnergyManagementBlock assigns EV chargers to either PV or grid and controls a
Micro_grid switch.
%
% Inputs:
%   P_pv    - Solar power (in Watts)
%   constant- A value from 0 to 5 indicating active EV chargers.
%   Soc     - Battery State-of-Charge (in percentage)
%
% Outputs:
%   For each EV charger i (1 to 5):
%       ev_pv_i   - 1 if charger i is connected to the PV bus, 0 otherwise.
%       ev_grid_i - 1 if charger i is connected to the grid, 0 otherwise.
%   Micro_grid  - 1 if the Microgrid is switched on, 0 otherwise.
%
% Conditions:
%   1. Initially, Micro_grid is off.
%   2. If constant == 0:
%        - Check the battery SOC.
%        - If SOC is 80% or above, turn on Micro_grid.
%        - All EV charger outputs are off.
%   3. If constant > 0:
%        - Convert P_pv from Watts to kW and add an offset of 30.
%        - Each EV charger is rated at 150 kW.
%        - Assign as many chargers as possible to PV without exceeding
available PV power.
%        - The remaining active chargers are fed from the grid.
%        - Micro_grid remains off.

% Initialize EV charger outputs to off (0)
ev_pv   = zeros(1,5);
ev_grid = zeros(1,5);

% Default: Micro_grid is off.
Micro_grid = 0;
```

```
if constant == 0
    % No EV chargers active.
    % Check battery SOC: if SOC is 80% or above, turn on Micro_grid.
    if Soc >= 80
        Micro_grid = 1;
    end
    % EV charger outputs remain off.
else
    % EV chargers are active.
    % Convert solar power from W to kW with an additional 50 kW offset.
    P_pv_kW = (P_pv / 1000) + 50;

    % Each charger requires 150 kW.
    % Determine how many chargers can be powered by PV.
    maxPVChargers = floor(P_pv_kW / 150);

    % Loop through each charger (1 to 5)
    for i = 1:5
        if i <= constant
            % Active charger.
            if i <= maxPVChargers
                % Assign charger to PV bus.
                ev_pv(i) = 1;
                ev_grid(i) = 0;
            else
                % Not enough PV power: assign charger to grid.
                ev_pv(i) = 0;
                ev_grid(i) = 1;
            end
        else
            % Inactive charger: both outputs remain off.
            ev_pv(i) = 0;
            ev_grid(i) = 0;
        end
    end
    % In this mode, Micro_grid remains off.
end

% Introduce a delay of 0.1 seconds (100 milliseconds)
%pause(0.1);

% Map array values to individual output variables
ev_pv_1   = ev_pv(1);
ev_grid_1 = ev_grid(1);
ev_pv_2   = ev_pv(2);
ev_grid_2 = ev_grid(2);
ev_pv_3   = ev_pv(3);
ev_grid_3 = ev_grid(3);
ev_pv_4   = ev_pv(4);
ev_grid_4 = ev_grid(4);
ev_pv_5   = ev_pv(5);
ev_grid_5 = ev_grid(5);

end
```

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